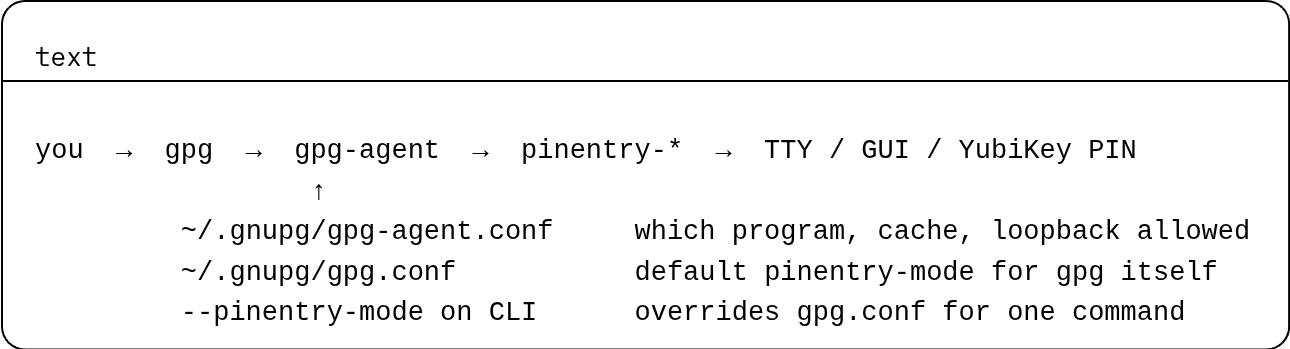


Pinentry is two layers: **which binary draws the box**, and **how gpg is allowed to talk to it**. Mixing those up is what locked your YubiKey PIN.

The stack



- **Program** (`pinentry-program`): `gtk`, `qt`, `gnome3`, `curses`, `tty`.
- **Mode** (`--pinentry-mode`): `ask` | `loopback` | `cancel` | `error` .
Loopback **does not use** the program. The agent asks *gpg* for the secret instead.

Programs (frontends)

Binary	UI	Best for	Notes
<code>pinentry-tty</code>	raw line on the TTY	SSH, scripts, no ncurses	Smallest surface; needs a real TTY
<code>pinentry-curses</code>	ncurses dialog	tmux/screen, SSH	Needs <code>GPG_TTY</code> ; fights stdin pipes
<code>pinentry-gtk-2</code>	GTK window	XFCE/MATE	Can fall back to curses
<code>pinentry-qt / qt5</code>	Qt window	KDE / any X	Quality bar, repeat PIN
<code>pinentry-gnome3</code>	GNOME System Prompter	GNOME	Needs <code>gnome-keyring D-Bus</code> ; can cache in <code>libsecret</code>
<code>pinentry-emacs</code>	Emacs minibuffer	Emacs only	Extra socket setup

GUI flavors often grab the keyboard (`grab`) so other windows cannot snoop. `no-grab` in `gpg-agent.conf` turns that off (weaker). GNOME/GTK/Qt can store secrets in `libsecret` unless you disable the external cache. TTY and curses do not.

On Gentoo the package is `app-crypt/pinentry`. USE flags pick binaries: `ncurses`, `gtk` (`gnome3`), `qt6`, `emacs`, `keyring` (`libsecret`), `X`, `wayland`. `/usr/bin/pinentry` is a symlink managed by:

Bash

```
eselect pinentry list
eselect pinentry set pinentry-curses
ls -l /usr/bin/pinentry
```

That symlink is only the **default**. `gpg-agent.conf` wins if it names a full path.

Files that matter

`~/.gnupg/gpg-agent.conf` — agent + pinentry program:

text

```
pinentry-program /usr/bin/pinentry-curses

# loopback is allowed by default on current GnuPG; pin it either way
allow-loopback-pinentry
# no-allow-loopback-pinentry

default-cache-ttl 0
max-cache-ttl 0

# do not let GNOME Keyring / libsecret remember the PIN
no-allow-external-cache

# YubiKey / SSH via gpg-agent (optional)
# enable-ssh-support
```

Reload after edits:

Bash

```
gpg-connect-agent reloadagent /bye
# or
gpgconf --kill gpg-agent
```

~/**.gnupg/gpg.conf** — default *mode* for every `gpg` invocation:

```
text
```

```
# dangerous for YubiKey + pipes – leave this unset
# pinentry-mode loopback
```

Keep loopback off globally. Pass it only on old `--symmetric` commands.

Environment the agent and curses/tty pinentry need:

```
Bash
```

```
export GPG_TTY=$(tty)
gpg-connect-agent updatestartuptty /bye
```

Without `GPG_TTY`, curses pinentry dies with `Inappropriate ioctl for device`. After you attach to `tmux` or `dtach`, run `updatestartuptty` again so the agent still knows the TTY.

Modes (again, in config terms)

Mode	Set where	What happens
ask	<code>--pinentry-mode ask</code> or default	Agent launches <code>pinentry-program</code>
loopback	CLI or <code>gpg.conf</code>	No pinentry binary; <code>gpg</code> reads the secret
cancel	CLI	Immediate cancel (use with <code>--list-packets</code>)
error	CLI	Pretend pinentry is missing

`--passphrase`, `--passphrase-fd`, and `--passphrase-file` only work with loopback (GnuPG ≥ 2.1). If loopback is forbidden (`no-allow-loopback-pinentry` or an old agent), you get something like `pinentry-mode loopback failed: Not supported / Forbidden`

you get `setting pinentry mode loopback failed: not supported / forbidden`.
Current default is **allow**.

Loopback + ciphertext on **stdin** is what produced your empty PIN. Curses pinentry on the same TTY as a pipe can steal stdin the same way. GUI pinentry is a separate window, so ... |
`gpg --decrypt --pinentry-mode ask` can work. File + ask is still safer.

Configurations that match your setup

YubiKey decrypt (what you want day to day)

text

```
# gpg-agent.conf
pinentry-program /usr/bin/pinentry-qt
# or pinentry-curses if you live in a terminal
allow-loopback-pinentry
default-cache-ttl 0
max-cache-ttl 0
no-allow-external-cache
```

Bash

```
export GPG_TTY=$(tty)
gpg --decrypt --pinentry-mode ask file.gpg
```

Zero TTL means every decrypt asks for the User PIN again. Raise `default-cache-ttl` (seconds) if you want “PIN once, then only touch for a while.” Touch policy is on the card (`ykman openpgp keys set-touch enc on`), not pinentry.

SSH session / no DISPLAY

text

```
pinentry-program /usr/bin/pinentry-curses
```

or `pinentry-tty`. GUI pinentry will fail over SSH unless you forward X/Wayland.

Old symmetric blob scripts

Do not change the agent default. On that one command only:

```
Bash
```

```
gpg --decrypt --pinentry-mode loopback --no-symkey-cache
```

Inspect packets without hitting the card

```
Bash
```

```
gpg --list-packets --pinentry-mode cancel
```

Emacs

allow-emacs-pinentry in gpg-agent.conf, eselect pinentry set pinentry-emacs, and (setq epa-pinentry-mode 'loopback) inside Emacs.

Cache knobs people confuse with pinentry

Option	Effect
default-cache-ttl / max-cache-ttl	How long the agent remembers a passphrase/PIN
--no-symkey-cache	Only for --symmetric ; ignore for YubiKey
no-allow-external-cache	Block libsecret / GNOME Keyring from storing the PIN
ignore-cache-for-signing	Always prompt for sign even if cached

A cached **wrong** PIN is why `gpgconf --kill gpg-agent` fixed odd “bad password” cases.

Suggested Gentoo set for your rclone scripts

```
Bash
```

```
# build the binaries you might switch between
echo 'app-crypt/pinentry ncurses gtk qt6' >> /etc/portage/package.use
emerge -av app-crypt/pinentry

eselect pinentry set pinentry-curses    # default symlink
```

text

```
# ~/.gnupg/gpg-agent.conf
pinentry-program /usr/bin/pinentry-curses
allow-loopback-pinentry
default-cache-ttl 0
max-cache-ttl 0
no-allow-external-cache
```

Leave pinentry-mode **out** of gpg.conf. Scripts: --pinentry-mode ask + temp file + GPG TTY That is the configuration that already decrypted with PIN + touch